

COUMADIN ANTICOAGULATION CLINIC

Department of Outpatient Pharmacy Services

1847 Cedar Ridge Boulevard · Suite 320 · Boston, MA 02114

Phone: (617) 555-2890 · Fax: (617) 555-2891

Date: September 18, 2019

Patient Name: Echevarría, Ester
Date of Birth: June 20, 1927 (Age 92)
Medical Record #: MRN-847261-EE
Indication: Atrial Fibrillation (Chronic)
Target INR: 2.0 – 3.0
Referring Physician: Dr. Marcus T. Greenfield, MD – Cardiology

ANTICOAGULATION MONITORING SUMMARY

This letter provides a comprehensive summary of Ms. Echevarría's warfarin sodium anticoagulation therapy over the past six months (March 2019 – September 2019). The patient is a 92-year-old female with chronic atrial fibrillation requiring long-term oral anticoagulation for stroke prevention.

HIGH-RISK PATIENT: Advanced age (92 years), concurrent chemotherapy (FOLFOX for recurrent rectal polyp), falls risk secondary to Alzheimer's disease, chronic kidney disease stage 1, polypharmacy.

CLINICAL BACKGROUND

Ms. Echevarría has been maintained on warfarin sodium 5 mg daily for chronic atrial fibrillation. She has a complex medical history including Alzheimer's disease, type 2 diabetes mellitus with associated retinopathy and renal disease, osteoarthritis, anemia requiring epoetin alfa, and recurrent rectal polyps currently undergoing FOLFOX chemotherapy (oxaliplatin plus leucovorin). Additional cardiac interventions include prior electrical cardioversion and catheter ablation.

Her concurrent medications include verapamil hydrochloride 40 mg, digoxin 0.125 mg, simvastatin 10 mg, insulin (Humulin 70/30), galantamine 4 mg for Alzheimer's disease, and intermittent antibiotics. The patient also undergoes renal dialysis. This polypharmacy, combined with chemotherapy and advanced age, significantly increases her bleeding risk and necessitates close INR monitoring.

INR MONITORING RECORD (MARCH – SEPTEMBER 2019)

Date	INR Value	Warfarin Dose	Clinical Notes
March 12, 2019	2.3	5 mg daily	Therapeutic range; continue current dose
March 26, 2019	2.7	5 mg daily	Therapeutic; no dose change
April 15, 2019	3.4	Reduced to 4 mg daily	Supratherapeutic; dose reduced. FOLFOX cycle initiated
April 22, 2019	2.1	4 mg daily	Therapeutic; dose adjustment effective
May 6, 2019	1.8	Increased to 5 mg daily	Subtherapeutic; dose increased
May 20, 2019	Held	Held × 3 days	FOLFOX cycle 2; warfarin held per oncology protocol
May 27, 2019	2.5	Resumed 5 mg daily	Resumed post-chemo; therapeutic
June 10, 2019	2.9	5 mg daily	Therapeutic; patient reports easy bruising on forearms
June 24, 2019	Held	Held × 3 days	FOLFOX cycle 3; warfarin held per protocol
July 1, 2019	2.2	Resumed 5 mg daily	Resumed post-chemo; therapeutic
July 15, 2019	3.6	Reduced to 4 mg daily	Supratherapeutic; recent amoxicillin/clavulanate course (sinusitis). Patient reports gum bleeding when brushing teeth
July 29, 2019	2.4	4.5 mg daily (alternating)	Therapeutic; alternating 5 mg / 4 mg schedule initiated
August 12, 2019	Held	Held × 3 days	FOLFOX cycle 4; warfarin held
August 19, 2019	2.6	Resumed alternating dose	Therapeutic post-chemo
September 2, 2019	2.8	Alternating 5 mg / 4 mg	Therapeutic; patient stable
September 16, 2019	2.5	Alternating 5 mg / 4 mg	Therapeutic; Continue current regimen

BLEEDING AND ADVERSE EVENTS

Over the six-month monitoring period, Ms. Echevarría has experienced the following bleeding-related events:

- **Easy bruising:** Reported on June 10, 2019. Patient noted bilateral forearm ecchymoses consistent with minor trauma. INR was 2.9 (therapeutic). No intervention required beyond education regarding fall prevention and protective measures.
- **Gingival bleeding:** Reported on July 15, 2019 during supratherapeutic INR of 3.6. Patient experienced mild gum bleeding during tooth brushing. Warfarin dose reduced from 5 mg to 4 mg daily. Bleeding resolved within 48 hours. Patient instructed to use soft-bristled toothbrush.
- **No major bleeding events:** No episodes of gastrointestinal bleeding, hematuria, intracranial hemorrhage, or other major bleeding requiring hospitalization or transfusion during this monitoring period.

DOSE ADJUSTMENTS AND MANAGEMENT CHALLENGES

Management of Ms. Echevarría's anticoagulation has been complicated by multiple factors:

Chemotherapy Interactions

FOLFOX chemotherapy (oxaliplatin and leucovorin) cycles required planned warfarin interruption for 3 days surrounding each infusion (May 20, June 24, August 12) to minimize bleeding risk during chemotherapy-induced thrombocytopenia. INR monitoring was intensified to every 7 days during chemotherapy cycles, with resumption of warfarin coordinated with the oncology team. Post-chemotherapy INR values returned to therapeutic range within 5-7 days of warfarin resumption.

Antibiotic Interactions

Amoxicillin/clavulanate prescribed for acute bacterial sinusitis (mid-July) resulted in INR elevation to 3.6 due to alteration of vitamin K-producing intestinal flora. Warfarin dose was reduced proactively, and INR returned to therapeutic range within one week.

Advanced Age and Polypharmacy

At 92 years of age, Ms. Echevarría demonstrates increased sensitivity to warfarin. Her concurrent medications—particularly verapamil, digoxin, simvastatin, and intermittent antibiotics—contribute to potential drug-drug interactions requiring vigilant monitoring. Alternating dose schedule (5 mg / 4 mg on alternating days) has provided more stable INR control since late July.

Falls Risk and Cognitive Impairment

The patient's Alzheimer's disease (currently managed with galantamine 4 mg) and osteoarthritis increase her falls risk. Family caregivers have been counseled extensively regarding fall prevention strategies, home safety modifications, and signs of intracranial bleeding. Despite these risks, the cardiology team and anticoagulation clinic agree that the stroke prevention benefit of warfarin outweighs bleeding risk in this patient with chronic atrial fibrillation.

CURRENT ANTICOAGULATION PLAN

Current Regimen

Warfarin Sodium 5 mg and 4 mg oral tablets, alternating daily:

- Monday, Wednesday, Friday, Sunday: 5 mg
- Tuesday, Thursday, Saturday: 4 mg
- Weekly dose: 31 mg (average 4.4 mg/day)

Monitoring Schedule

- **Next INR check:** September 30, 2019 (2 weeks)
- **Ongoing frequency:** Every 2-3 weeks if INR remains stable within therapeutic range (2.0-3.0)
- **Intensified monitoring:** Weekly INR checks during any chemotherapy cycles, new antibiotic courses, acute illness, or dose adjustments

Dietary and Lifestyle Counseling

Patient and family caregivers have received ongoing education regarding:

- Consistent dietary intake of vitamin K-containing foods (green leafy vegetables)
- Avoidance of over-the-counter NSAIDs (patient already takes naproxen sodium 220 mg as prescribed; acetaminophen 325 mg preferred for breakthrough pain)
- Fall prevention strategies including removal of trip hazards, adequate lighting, use of assistive devices
- Signs and symptoms requiring immediate medical attention (severe headache, vision changes, chest pain, black tarry stools, frank blood in urine or stool, uncontrolled bleeding)
- Medication adherence strategies including weekly pill organizer with caregiver supervision

Coordination of Care

Ongoing communication maintained with:

- Dr. Marcus T. Greenfield (Cardiology) – primary anticoagulation prescriber
- Dr. Helena Rasmussen (Oncology) – coordination of warfarin holds during FOLFOX cycles
- Dr. William Chen (Primary Care/Geriatrics) – overall medical management
- Nephrology team – coordination with dialysis schedule
- Home health nursing – medication administration oversight and INR specimen collection

SUMMARY AND RECOMMENDATIONS

Ms. Echevarría's anticoagulation therapy has been moderately stable over the past six months, with most INR values falling within the therapeutic range of 2.0-3.0. The alternating-dose schedule (5 mg/4 mg) implemented in late July has improved INR stability. Planned warfarin interruptions during chemotherapy cycles have been managed safely with appropriate monitoring.

The patient remains at high risk for both thrombotic and hemorrhagic complications due to advanced age, falls risk, cognitive impairment, concurrent chemotherapy, chronic kidney disease, and polypharmacy. Minor bleeding events (easy bruising, gingival bleeding) have occurred but no major hemorrhages. The risk-benefit analysis continues to favor anticoagulation for stroke prevention in the setting of chronic atrial fibrillation.

Recommendations going forward:

- Continue warfarin sodium alternating dose schedule (5 mg/4 mg)
- INR monitoring every 2-3 weeks if stable; weekly during chemotherapy or acute illness
- Continue close coordination with oncology for planned warfarin holds during chemotherapy
- Maintain caregiver education regarding fall prevention and bleeding precautions
- Consider home INR monitoring device if patient/family demonstrates capability and interest
- Reassess risk-benefit of continued anticoagulation with cardiology team every 6 months given advancing age and progressive cognitive decline

The patient and her family caregivers understand the anticoagulation plan and have been provided with written instructions, emergency contact numbers, and warfarin wallet card. They have been instructed to contact the Coumadin Clinic immediately for any bleeding concerns, falls with head trauma, or inability to obtain scheduled INR testing.

If you have any questions regarding this patient's anticoagulation management, please contact the Coumadin Clinic at (617) 555-2890.

Katherine R. Morrison, PharmD, BCPS, CACP

Clinical Pharmacist, Anticoagulation Services

Board Certified Pharmacotherapy Specialist

Certified Anticoagulation Care Provider

Massachusetts Pharmacist License #PH-48291

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